

Incentives, risk, and ambiguity preferences bias perceptual confidence judgements

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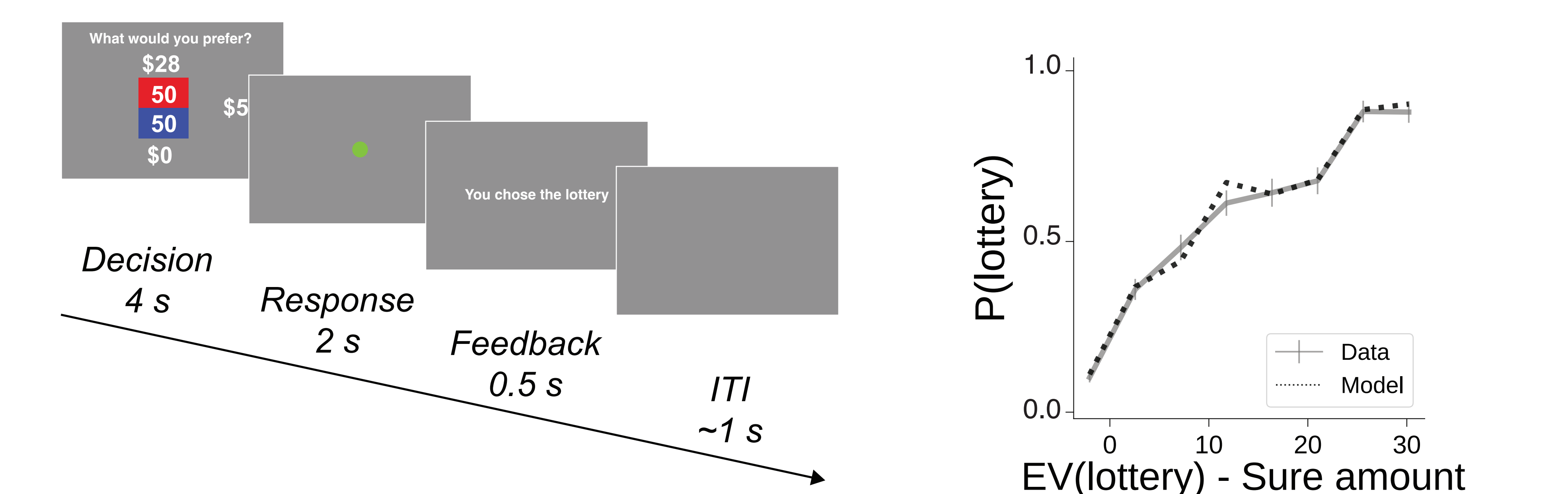
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Background

- Decision confidence is thought to represent an estimate of the probability of making correct choices. However, prior work shows that monetary incentivization may induce biases that alter confidence judgements¹.
- If incentivized confidence reports are perceived as lotteries, then they may reflect utility maximization, rather than a pure decision confidence computation.
- Here, we ask whether economic preferences, such as risk² and ambiguity preferences, might explain individual differences in incentive-induced bias in confidence judgement.
- We model decision confidence as a noisy estimate of decision reliability using the CASANDRE³ framework. We test whether incentives and economic preferences influence CASANDRE parameters differently, specifically confidence criterion and confidence sensitivity (meta-uncertainty).
- Participants recruited online on Prolific and Cloud Research (N = 343) completed a risk and ambiguity preference task and a visual orientation discrimination task with simultaneous confidence reporting.

Risk and Ambiguity Decision-making Task



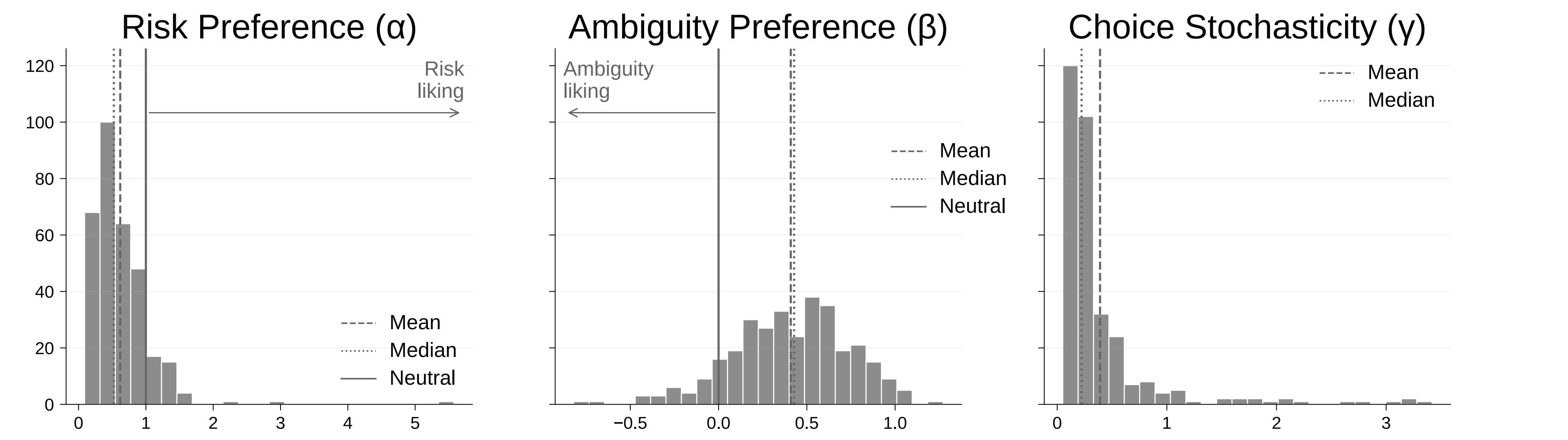
Utility Model on Risky Decision:

Subjective Value:

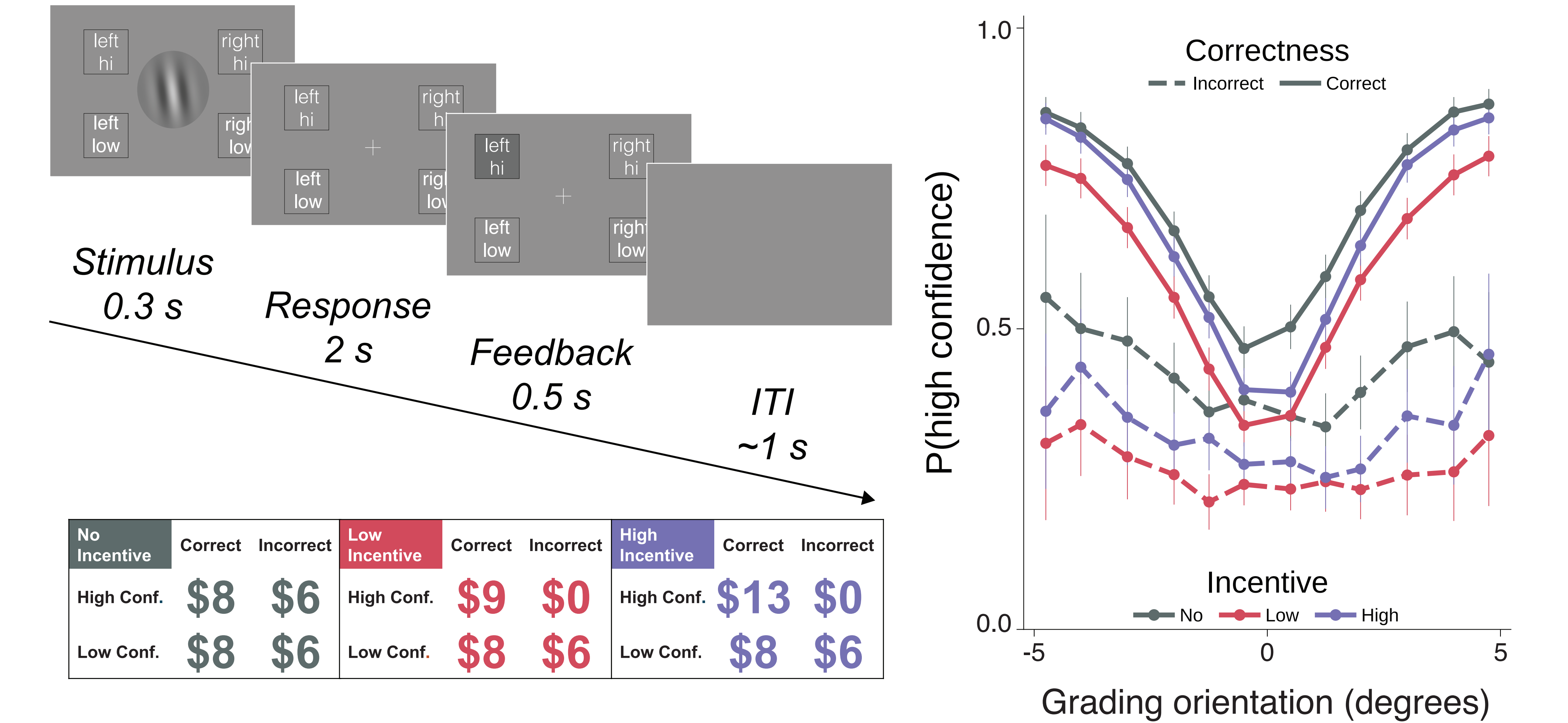
$$SV_{option} = [p - \beta \frac{A}{2}] v^\alpha$$

Choice Rule:

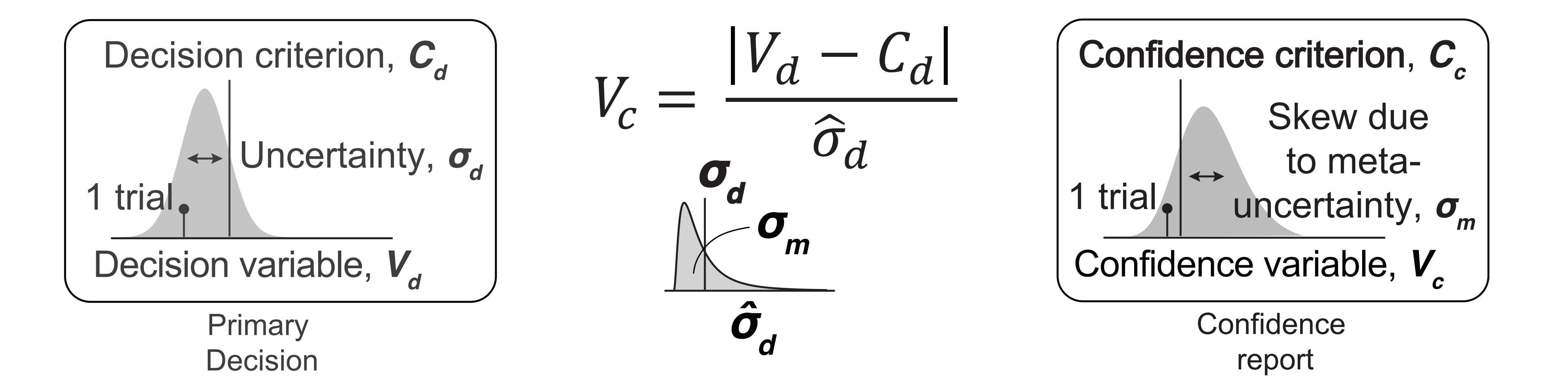
$$P_{lottery} = \frac{1}{1 + e^{-\gamma(SV_{option} - SV_{alt.})}}$$



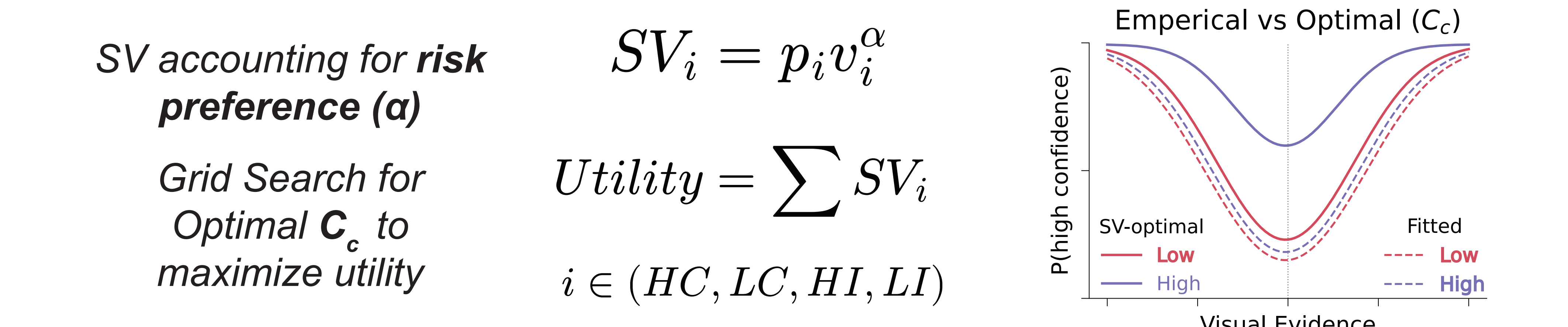
Confidence in Perceptual Decision-making Task



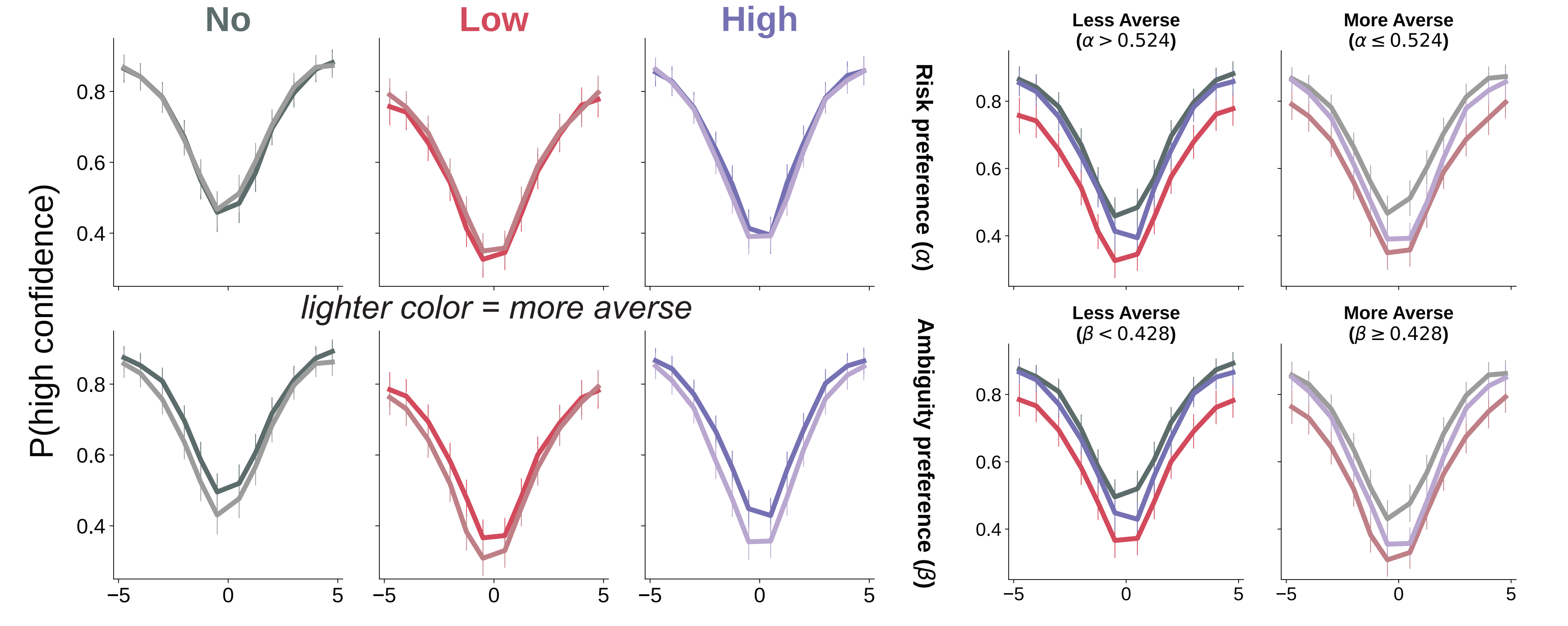
CASANDRE Model of Confidence



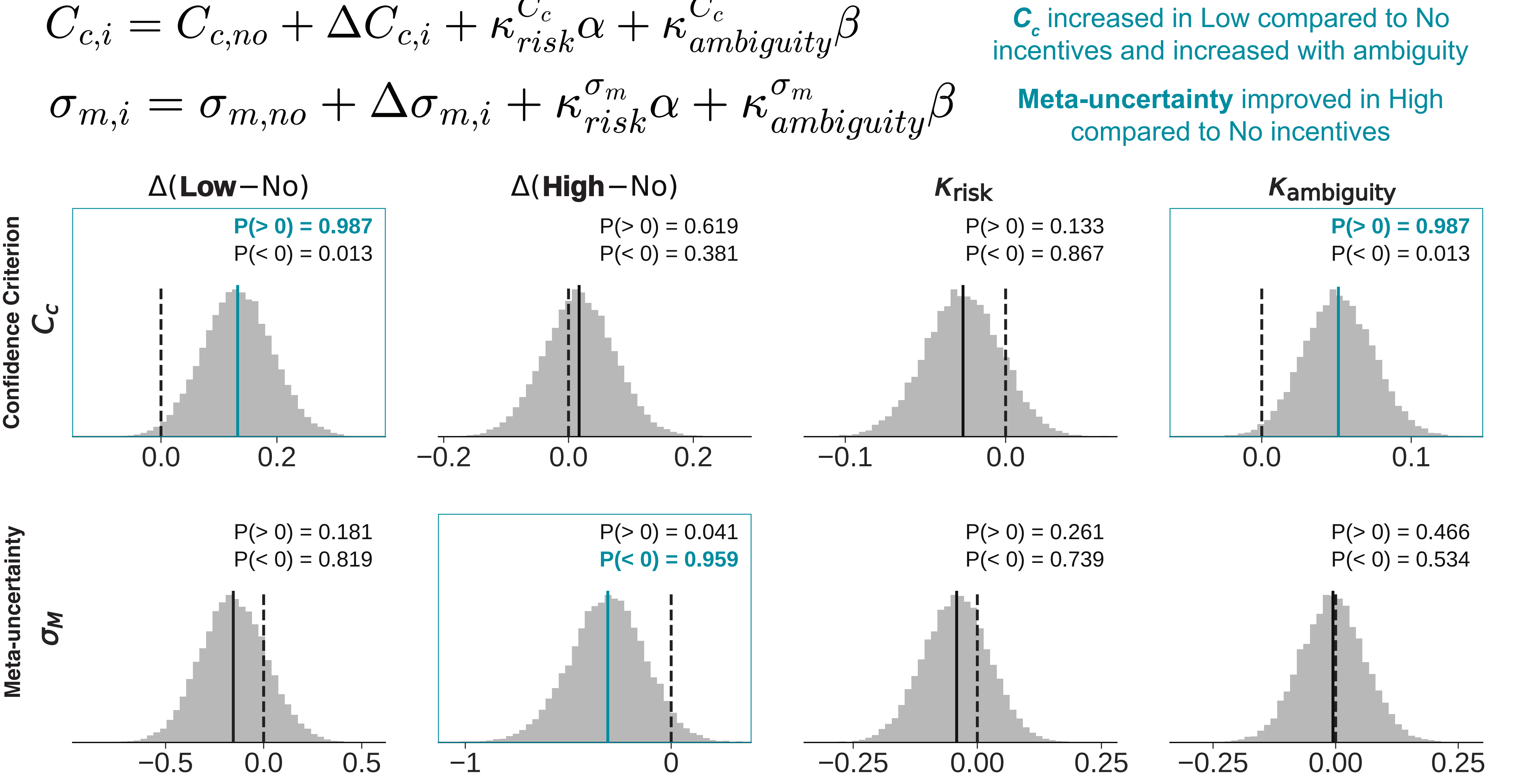
Simulation: Utility Maximizing Ideal Observer



Confidence Report by Risk and Ambiguity Preference



Effects of incentive, risk and ambiguity on C_c and σ_m



Risk and Ambiguity Intergrated Confidence Models

SV Gamble Model:

$$SV_j = p_{sv} v_{cor}^\alpha + (1 - p_{sv}) v_{incor}^\alpha$$
$$p_{sv} = p_{cor} - \beta \frac{Q_{95}(V_c) - Q_5(V_c)}{2}$$

Mean-variance Model

$$SV_j = \mu_j + \epsilon_{var} \sigma_j^2$$
$$\mu_j = p_{cor} v_{cor} + (1 - p_{cor}) v_{incor}$$
$$\sigma_j^2 = p_{cor} (1 - p_{cor}) (v_{cor} - v_{incor})^2$$

CASANDRE with β bias

$$\mathcal{D}(S) = 1 - |P(V_d > C_d | S) - P(V_d < C_d | S)|$$
$$V_c^*(S) = \frac{|V_d - C_d|}{\hat{\sigma}_d \exp[\beta \mathcal{D}(S)]} = V_c \exp[-\beta \mathcal{D}(S)]$$

Conclusions

- Monetary incentives impact confidence reports: Incentives improve confidence sensitivity (by reducing meta-uncertainty), and also bias confidence levels (by changing confidence criteria).
- The effect of incentives on confidence criteria is sensitive to the magnitude of the reward (more conservative for lower incentives and more liberal for higher incentives), and to individual economic preferences.
- Surprisingly, ambiguity preference, rather than risk, has a greater effect in confidence judgement. Future work will focus on modeling the mechanisms of this effect.